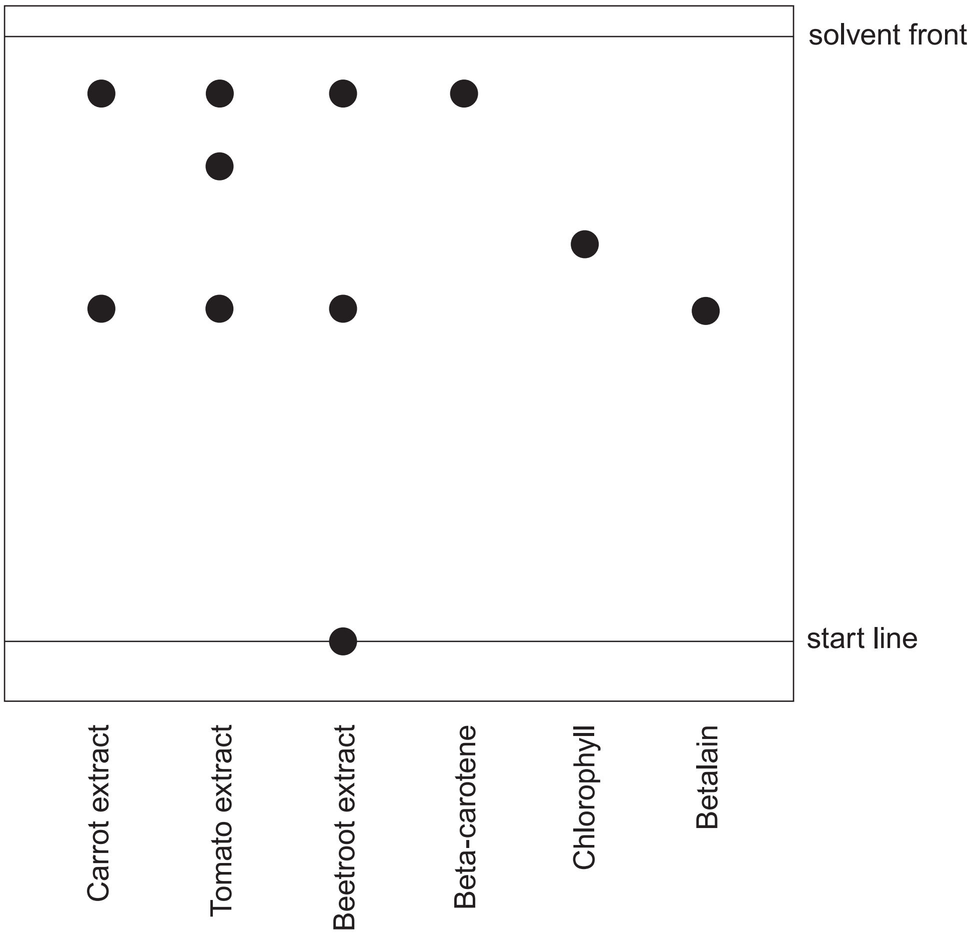
Examiner only

3. A student investigates the pigments found in some fruit and vegetables.

She obtains some coloured extracts from carrots, tomatoes and beetroot.

She places a spot of each extract on chromatography paper, along with spots of the three pigments beta-carotene, chlorophyll and betalain.

The diagram shows the chromatogram at the end of the experiment.



(a) Give the names of **all** the substances which are **not** mixtures. Give a reason for your choice. [1]

Substances

....

Reason

....

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(b) Put a tick (✓) in the boxes next to the **two** conclusions that can be drawn from the chromatogram. [2]

chlorophyll is not present in carrot, tomato or beetroot extracts ☐

beta-carotene is present in carrot extract but not present in tomato extract ☐

both beta-carotene and betalain are present in beetroot extract ☐

betalain is present in tomato extract but not present in carrot extract ☐

both carrot and beetroot extracts contain a pigment other than beta-carotene, chlorophyll and betalain ☐

(c) One of the pigments present in the carrot extract has travelled 4.4 cm above the start line.

The solvent front has travelled 8.0 cm. Calculate the R_f value of the pigment using the following equation. [2]

$$R_f = \frac{\text{distance travelled by pigment}}{\text{distance travelled by solvent front}}$$

R_f = ...

(d) Give the reason why there is a spot remaining on the start line in the chromatogram for beetroot. [1]

....

....

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